



Mini-design Project: Design and Analysis of a Simplified Impact-Resistant Steel Truss

School of Civil and Environmental Engineering

Term 2, 2026

ENGG1300 Engineering Mechanics

Project Brief and Guidelines

Many civil, structural, mining, and mechanical systems can be idealised as simple truss structures. Examples include scaffolding, bridge side trusses, crane support frames, and elevated work platforms. Although these structures serve different purposes, their basic load transfer mechanisms can often be studied using the same mechanics principles.

In this assignment, you will analyse a simplified two-dimensional steel truss subjected to:

- a vertical static load; and
- a horizontal impact load.

The purpose of this assignment is to apply statics and dynamics knowledge to a realistic engineering problem. You will use free body diagrams, equilibrium equations, truss member force analysis, internal force diagrams, work-energy, and basic stress checks.

Submission requirements

Your final submission must be uploaded as **one PDF** report via Moodle, including all your hand-written calculation process and screenshots of another calculation method selected.

The report should be a maximum of **20 A4 pages** and should follow the formatting requirements below:

- 25 mm margins on all sides;
- 12 pt Times New Roman font;
- single or up to 1.25 line spacing;
- pages must be numbered;
- figures and tables must be clearly labelled and referred to in the text;
- equations and calculations must include appropriate units.

Your report should include:

1. cover page and contents;
2. project background and selected structure;
3. truss drawing and dimensions;
4. geometric property calculations;
5. static load analysis;
6. dynamic impact load analysis;
7. axial stress check;
8. critical member and threshold diameter calculation;
9. comparison between hand calculation and another calculation method;
10. short reflection;
11. references if possible.

Software or AI output alone is not sufficient. Hand calculations must be shown. If software or AI tools are used, they must be clearly acknowledged and checked.

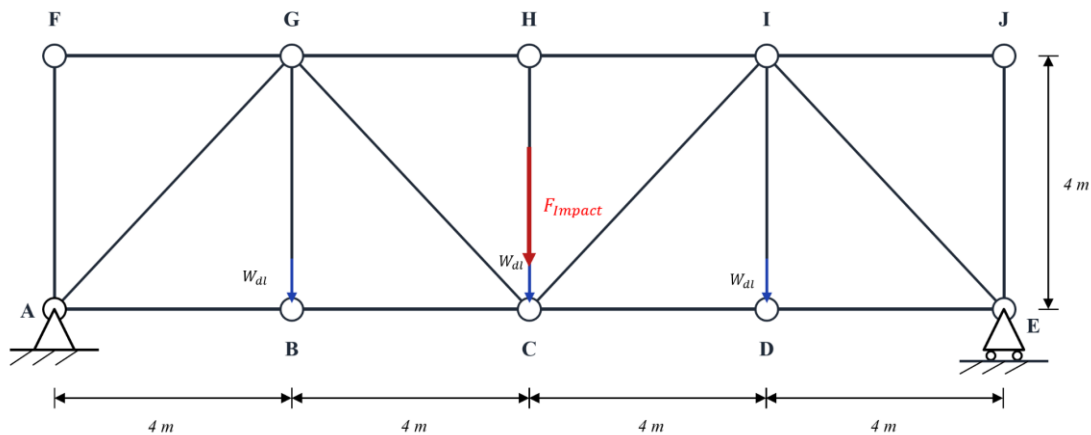
Suggested timeline

Week	Suggested Progress
Week 3	Familiarise yourself with the project and develop a clear understanding of the task requirements.
Week 5	Complete structure selection, Section 4.1 geometric properties and Section 4.2 static load analysis.
Week 8	Complete Section 4.3 dynamic impact load analysis and member force calculations.
Week 10	Complete Section 4.4 - 4.5 stress check, critical member identification, threshold diameter calculation; Section 5 two-method comparison, and report writing.
Week 11	Final report due by Friday 20:00 pm, 14th August.

1. Structure Selection

Demonstration Case: Pedestrian footbridge side truss.

A small pedestrian footbridge uses side trusses to support vertical pedestrian loads. The truss carries loads from pedestrians, the deck system, and small maintenance equipment. A dynamic impact caused by people jumping may accidentally act on the lower chord.



Simplified structure: 2D bridge side truss.

For this demonstration:

- Static loads (W_{dl}) are applied to selected lower deck joints, such as B, C and D, for the static analysis.
- The impact load is applied vertically or laterally at joint C along with the static loads, depending on the loading direction shown in the figure.
- The structure is supported by a pin support at A and a roller support at E.
- The main engineering question is whether the bridge side truss can transfer the deck load and resist the local accidental impact without excessive member stress.

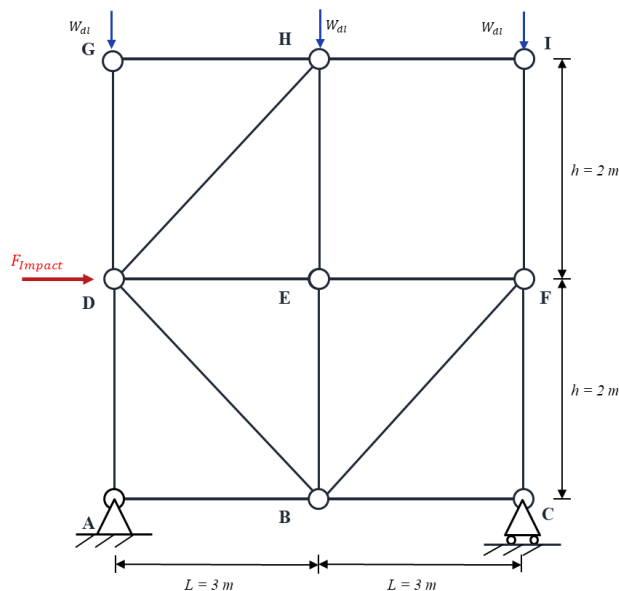
For your project, please select **ONE** of the following simplified engineering structures. Each option represents a different civil, structural, or mechanical engineering application, but all options can be idealised as a two-dimensional steel truss.

For your selected structure, you will conduct:

1. a **static analysis** under vertical static loads; and
2. a **dynamic analysis** under an equivalent impact load.

Option 1: Temporary construction scaffolding frame

A temporary scaffold is installed beside a building during maintenance work. The scaffold supports workers, tools, and small equipment. During site operation, a moving trolley or small site vehicle accidentally impacts the lower part of the scaffold. The scaffold must remain stable and transfer the loads safely to the ground.



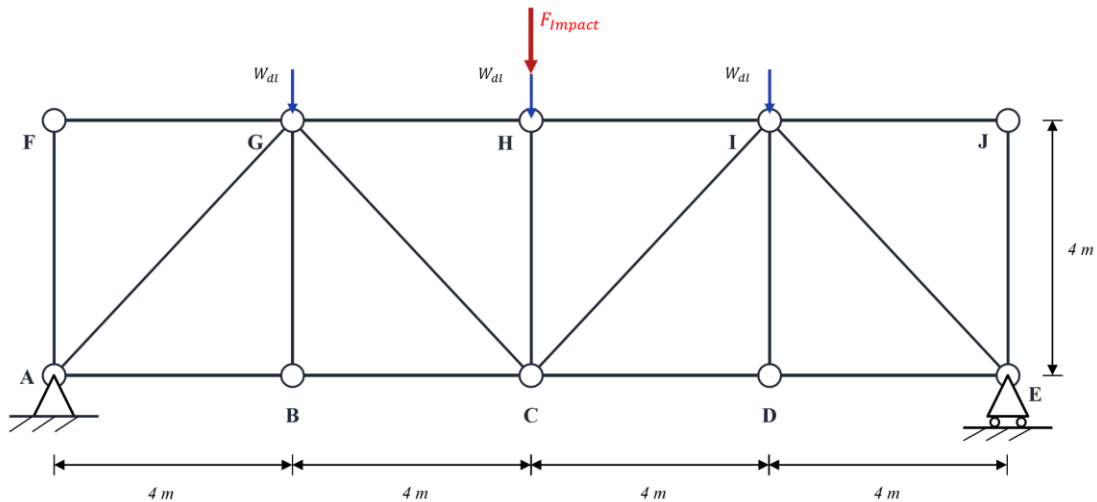
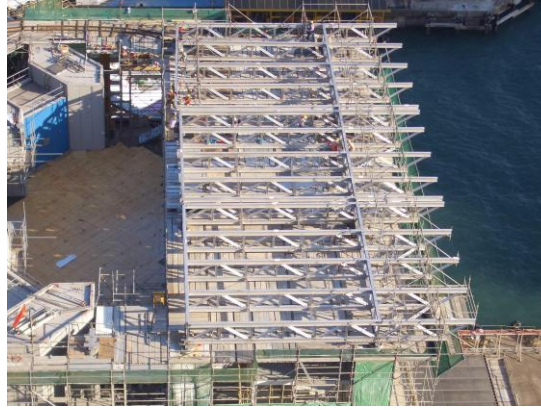
Simplified structure: two-storey 2D steel truss/scaffold frame.

For this option:

- Static loads are applied to the upper joints G, H and I for the static analysis.
- The impact load is applied horizontally at joint D for the dynamic analysis, along with the static loads.
- The impacting object accelerates horizontally from rest over a distance of 3 m at $a = 5.00 \text{ m/s}^2$ before hitting the scaffold.
- The structure is supported by a pin support at A and a roller support at C.
- The main engineering question is whether the scaffold members can safely carry the vertical loads and the accidental lateral impact force.

Option 2: Helipad side truss

A small, elevated helipad or service landing platform may use side trusses to support the platform deck. The structure carries vertical static loads from the platform, safety barriers, and equipment. During operation, a suspended component is not properly secured, falls over a short distance, and introduces a local vertical impact load to the platform deck.



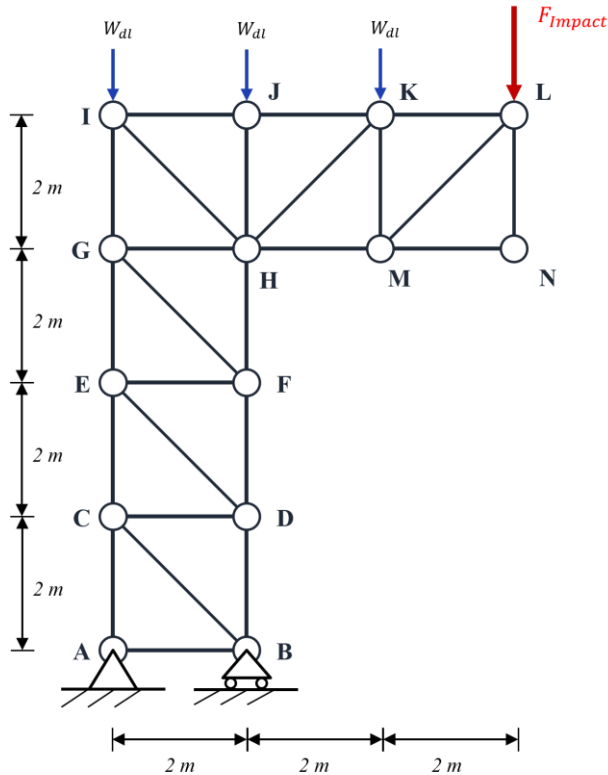
Simplified structure: 2D helipad side truss.

For this option:

- Static loads are applied to the upper or platform-level joints, such as G, H and I, for the static analysis.
- The impact load is applied vertically at joint H for the dynamic analysis, along with the static loads.
- The impacting object falls freely from rest over a vertical distance of 3 m before contacting the helipad side truss under gravitational acceleration of $g = 9.81 \text{ m/s}^2$.
- The structure is supported by a pin support at A and a roller support at E.
- The main engineering question is whether the truss can safely transfer both the regular platform load and the local dynamic landing/service impact to the supports.

Option 3: Small workshop crane support truss

A small workshop crane is supported by a steel truss frame. The frame carries vertical loads from the crane rail and suspended materials during regular operation. In an accidental event, a construction material or suspended component is not properly secured, falls over a short distance, and impacts the crane support at joint L. The truss must safely transfer both the regular vertical loads and the local vertical impact load to the supports.



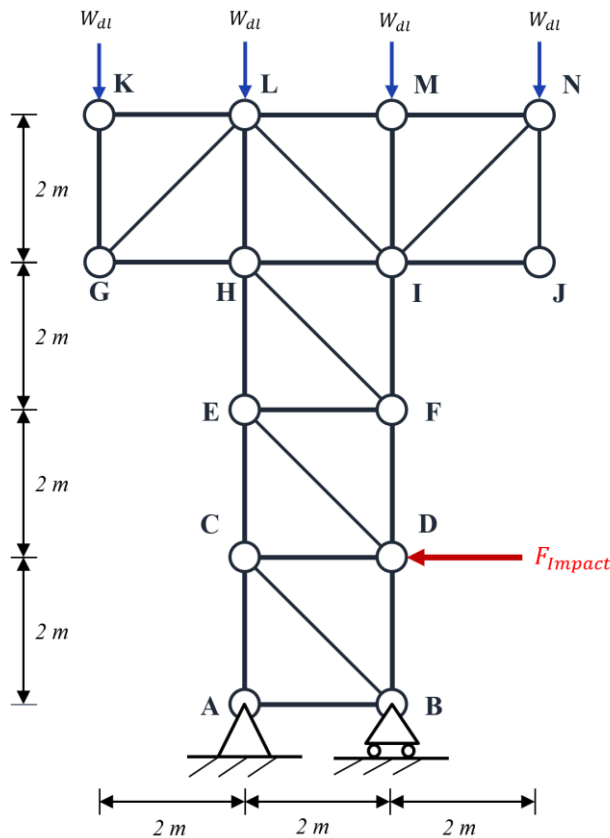
Simplified structure: two-storey crane support truss.

For this option:

- Static loads are applied to the top or crane-support joints, such as I, J and K, for the static analysis.
- The impact load is applied vertically downward at joint L for the dynamic analysis, along with the static loads.
- The impacting object falls freely from rest over a vertical distance of 3 m before contacting the crane support truss under gravitational acceleration of $g = 9.81 \text{ m/s}^2$.
- The structure is supported by a pin support at A and a roller support at B.
- The main engineering question is whether the crane support truss can safely carry the suspended load and resist accidental impact during crane operation.

Option 4: Elevated maintenance work platform

An elevated work platform supports one worker and equipment. The platform is braced by steel members. A moving cart or suspended component accidentally impacts one side of the lower frame. The structure must carry the vertical platform load and resist the impact load through its bracing system.



Simplified structure: braced platform frame.

For this option:

- Static loads are applied to the platform-level joints, such as K, L, M and N, for the static analysis.
- The impact load is applied horizontally at joint D for the dynamic analysis, along with the static loads.
- The impacting object accelerates horizontally from rest over a distance of 3 m at $a = 5.00 \text{ m/s}^2$ before hitting the platform.
- The structure is supported by a pin support at A and a roller support at B.
- The main engineering question is whether the platform frame can transfer the vertical load and resist the lateral impact through the braced members.

2. Material and Member Properties

Use the following steel properties for all members:

Property	Value
Density (ρ)	7850 kg/m ³
Young's Modulus (E)	200 GPa
Allowable axial stress ($\sigma_{\text{allowable}}$)	80 MPa

The allowable stress is provided for educational purposes only. It is not intended to represent a complete Australian Standard design value.

All members are solid circular cross-section with diameter:

$$d = 30 \text{ mm}$$

You must calculate the following geometric properties:

- Member self-weight
- Cross sectional area
- Second moment of area

3. Loading Conditions

Each student must use the loading values assigned according to the first letter of their surname.

3.1 Static load case: static load

The static load is represented by the static load W_{dl} applied at each specified loading joint. Use the loading value assigned according to the first letter of your surname.

Surname group	Total static load W_{dl} (kN)
A-E	8
F-J	9
K-O	10
P-T	11
U-Z	12

3.2 Dynamic load case: impact load

The dynamic action is represented by an equivalent impact force, calculated using the following information.

Quantity	Value
Acceleration distance (s_a)	3.00 m
Effective stopping distance (s_s)	0.10 m
Vertical acceleration (g)	9.81 m/s ²
Horizontal acceleration (a)	5.00 m/s ²

Use the following surname-based vehicle or impact mass.

Surname group	Mass m (kg)
A-E	40
F-J	50
K-O	60
P-T	70
U-Z	80

4. Required Calculations

Your report must include the following calculations.

4.1 Geometric properties

For the members in your selected structure, calculate the following properties.

- Self-weight.
- Cross-sectional area.
- Second moment of area.

4.2 Static load analysis: free body diagram, support reactions and truss member forces

For the static load case, use the static load value assigned to your surname group in Section 3.1 and complete the static analysis for the **entire structure**.

4.3 Dynamic load analysis: free body diagram, support reactions and truss member forces

For the dynamic load case, use the impact mass assigned to your surname group in Section 3.2, together with the static loads.

You must first calculate the velocity immediately before impact using the given acceleration and acceleration distance. Then calculate the equivalent impact force using the stopping distance. Apply the impact force at the impact joint shown in your selected **entire structure**.

For **Section 4.2 & 4.3**, your submission must include:

1. a complete free body diagram of the full structure;
2. all support reactions;
3. the axial force in every truss member;
4. clear indication of whether each member is in tension or compression;
5. force directions shown on the relevant joint diagrams or member force diagrams;
6. sufficient calculation steps to show how the member forces were obtained.

You may use the method of joints, method of sections, or a combination of both. The final member-force results for both static and dynamic cases should be summarised in a table similar to the following format:

Member	Axial force under static load	Tension /Compression
AB		T / C
AC		T / C
...		...

4.4 Axial stress check

For each member, calculate the axial stress using:

$$\sigma = \frac{N}{A}$$

where:

- N = axial force in member;
- A = cross sectional area.

Then compare the calculated stresses with the allowable stress:

$$\sigma_{max} \leq \sigma_{allowable}$$

Identify all members that do NOT pass the simplified stress requirement under the static and dynamic load cases .

4.5 Critical member and threshold diameter

After completing the static and dynamic analyses, you must identify the **most critical member**. The most critical member is defined as: *the member with the largest absolute axial force among all members analysed under loading*.

Based on the requirement of the allowable stress, calculate the minimum diameter required for the most critical member to satisfy the allowable stress limit, for both static and dynamic cases respectively.

For the threshold diameter calculation, you only need to consider the member with the maximum absolute axial force. The threshold diameter should be calculated separately for the static and dynamic load cases, using the largest absolute axial force obtained in each case.

5. Requirement for Two Calculation Methods and Reflection

You must solve the main analysis using **hand calculation**, and check your results using another method, such as:

- Space Gass;
- MATLAB;
- AI-assisted calculation may be used only if the assumptions, equations, and checking process are clearly documented;
- Other structural analysis and design software.

You must include a short reflection of approximately 150-250 words discussing:

- whether the two methods agree;
- possible reasons for any difference;
- which assumptions were used in each method;
- what you learned from comparing the methods.

Marking Criteria

Criterion	Marks	Assessment standard
1. Geometry, structure selection and report setup	2	1 mark: correctly identifies the selected structure, surname group, assigned static load, assigned impact mass, load locations and support conditions. 1 mark: provides a clear and readable report setup with labelled figures, consistent notation and stated assumptions.
2. Geometric properties and material use	4	1 mark: correctly uses the given steel properties and member diameter. 1 mark: correctly calculates cross-sectional area. 1 mark: correctly calculates second moment of area. 1 mark: correctly calculates self-weight for at least one selected member with appropriate units.
3. Static support reactions	3	1 mark: draws a correct full-structure FBD under static loading. 1 mark: correctly calculates vertical support reactions. 1 mark: correctly identifies any horizontal support reaction or explains why it is zero.
4. Static truss member forces	5	1 mark: uses a suitable method of joints / sections with clear calculation steps. 2 marks: calculates axial forces for all members with generally correct equilibrium procedure. 1 mark: correctly identifies tension / compression for all members. 1 mark: presents static member-force results clearly in a table or labelled diagram.
5. Dynamic load analysis	5	1 mark: correctly selects the acceleration value and calculates the impact velocity using the 3 m acceleration distance. 1 mark: correctly calculates the equivalent impact force and applies it at the correct joint and in the correct direction.

		1 mark: draws a correct full-structure FBD and calculates support reactions under dynamic loading. 1 mark: calculates axial forces for all members with a generally correct equilibrium procedure. 1 mark: correctly indicates force directions and tension / compression states for dynamic member forces.
6. Axial stress check and critical member	4	2 marks: correctly calculates axial stresses for static case. 2 marks: correctly calculates axial stresses for dynamic case.
7. Threshold diameter	2	1 mark: correctly identifies the most critical member as the member with the maximum axial stress among the analysed members for static and dynamic cases. 1 mark: correctly calculates the threshold diameter using the stress requirement for static and dynamic cases.
8. Two-method comparison and reflection	4	1 mark: uses hand calculation and one appropriate additional method, such as spreadsheet, MATLAB, Python, truss software, or AI-assisted calculation with acknowledgement. 1 mark: compares support reactions from the two methods. 1 mark: compares axial forces from the two methods. 1 mark: provides a 150-250 word reflection on agreement, assumptions, possible differences, and what was learned.
9. Overall communication, units and presentation	1	1 mark: calculations are readable, units are consistent, tables / figures are clearly labelled, and the submission is professionally presented.
Total	30	